Six Handshakes to a Stranger

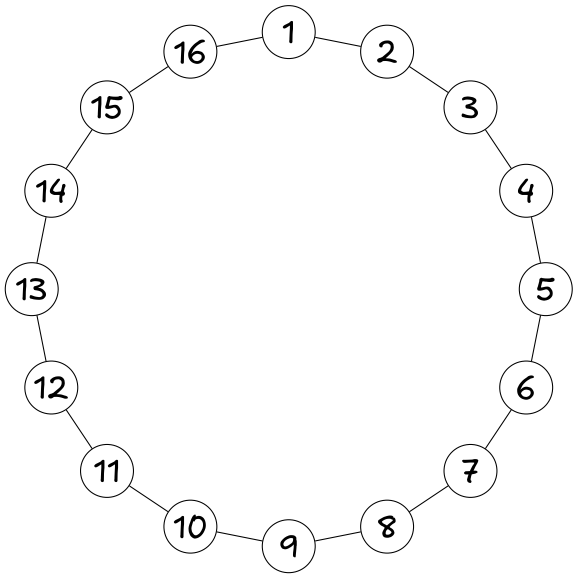
In 1967, a psychologist mailed a packet to a few hundred people in Omaha, Nebraska. The instructions: this packet must reach a stockbroker in Boston. You may not mail it to him directly. You may only hand it to someone you know personally on a first-name basis, and ask them to do the same.

Question 1: Write down your guess before reading on. How many hands did the packet pass through before it arrived? ________

And write down why you guessed that number.

Part 1: The town of Ringville

Sixteen people live in Ringville. They sit in a circle, and each person knows only the two people sitting next to them — nobody else.



Question 2: Person 1 wants to get a message to person 9. Trace the shortest chain of handshakes on the drawing. How many handshakes is it? ________

Now do this for everybody. Fill in the number of handshakes from person 1 to each other person.

To	2	3	4	5	6	7	8	9
Handshakes								

To	10	11	12	13	14	15	16
Handshakes							

Question 3: On average, how far away is a Ringville resident from person 1? ________

And how far away is the unluckiest pair — the two people who need the most handshakes of anyone in town? ________

Give these two numbers a name of your own. Which one would you quote to a newspaper, and which one to an engineer who has to guarantee delivery?

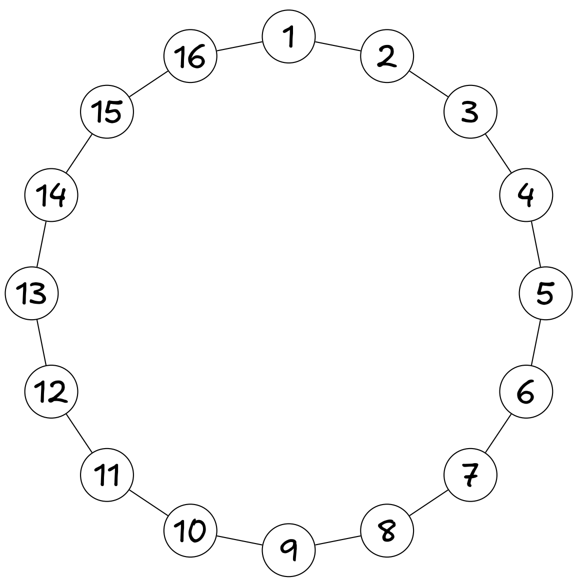
Question 4: Ringville grows. Now 32 people sit in the circle, still shaking hands only with their two neighbors. You do not need to redraw anything — just reason about it.

1. The unluckiest pair in the 32-person town is ________ handshakes apart.
2. If the town grows to 1,000 people? ________ To 8 billion? ________
3. In one sentence: how does the worst case grow as the town grows?

   ____

Ringville is a perfectly reasonable model of a village: your friends are the people near you. But the numbers you just wrote down say that a world wired this way could never deliver a packet from Omaha to Boston in a handful of steps. Something must be wrong with the model.

Question 5: Two people in Ringville go to college far away and each makes one distant friend back home. Add exactly two new connections to the town: 1–9 and 5–13. Draw them on the circle below.



Now redo the table of Question 2 for the new town.

To	2	3	4	5	6	7	8	9
Handshakes								

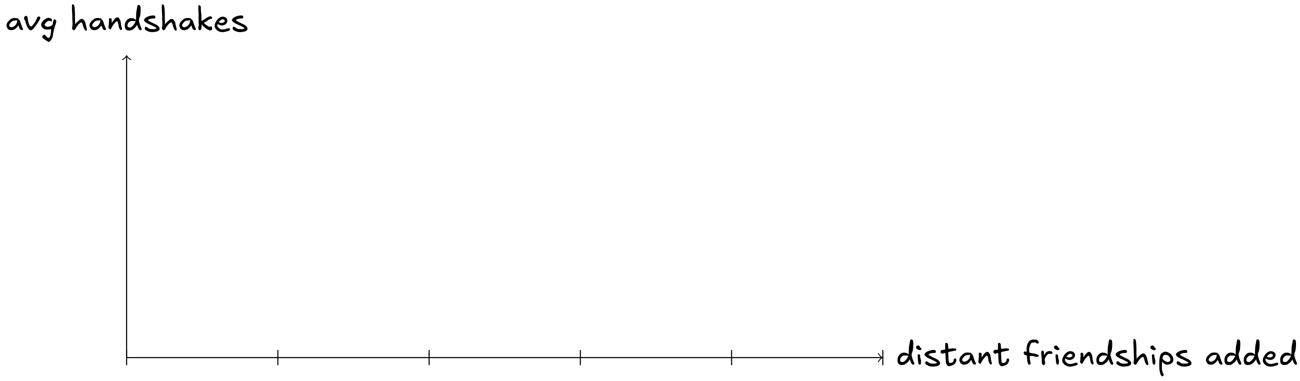
To	10	11	12	13	14	15	16
Handshakes							

Question 6: Fill in the comparison.

	Before (16 edges)	After (18 edges)
Average handshakes from person 1		
Unluckiest pair in town		

Two connections out of eighteen were changed — about 11% of the town's friendships. Was the effect on distance also about 11%? What did those two connections actually do?

Question 7: Predict, without computing: you add four more distant friendships. Does the average drop by as much again as it did the first time? Sketch a rough curve of "average handshakes" against "number of distant friendships added", from 0 to 16.



Question 8: Here is the catch. You are person 1 in the rewired town, and you are holding a packet for person 12. You cannot see the drawing — you only know who your own contacts are and where each of them sits in the circle. You hand the packet to one of them, and they face the same situation.

1. Play it out. Write the chain you would produce: $1 \rightarrow$ ________________
2. Is your chain the shortest one that exists? ________
3. So: are "a short chain exists" and "ordinary people can find a short chain" the same claim? Which one did the Omaha packet demonstrate?

____

Question 9: Suppose someone hands you a network of a billion people and asks for the distance from one person to everyone else. You do not need to do it — just write down the procedure you would follow, in the same way you filled in the table in Question 2. What do you do first, and what do you do at each step after that?

Question 10: Back to Omaha. Suppose every person knows 150 people, and — pretend for a moment — no two of your contacts share a contact.

Handshakes away	1	2	3	4
People you can reach				

At how many handshakes do you exceed 8 billion people? ________

Compare with the guess you wrote in Question 1. Was your guess too big or too small — and now that you have both Ringville and this calculation in front of you, which assumption in this last calculation is the one you should distrust?

____